

Boundary Conditions as Structural Selectors: A Formal Analysis of N_0 , Operational Constraints, and N_1 in Self-Optimizing Systems

A Contribution to the Theory of Axiomatic Necessity (TNA)

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Abstract

This paper formalizes a critical distinction within the Theory of Axiomatic Necessity (TNA): the difference between a system's internal dynamics (N_0), the operational constraints that condition its solution space (boundary conditions), and the selecting instance that imposes those constraints (N_1). Through analogies drawn from physics (Laplace equation, Euler-Bernoulli equation, Manning equation), hydrodynamics (channel flow), wave mechanics, and artificial intelligence (LLM system prompts), we demonstrate that systems with identical internal dynamics can produce radically different trajectories when their boundary conditions vary. We identify and respond to materialist objections, showing that the complete reduction of boundary condition selection to N_0 leads to either infinite regress or ontological collapse. The central thesis is that the identity of complex systems —biological, cognitive, or social— does not reside in their constituent matter (which is replaceable) but in the realizability structure that persists through material turnover.

1. Introduction: The Problem of Local Closure

The Theory of Axiomatic Necessity (TNA) has developed a formal framework for analyzing the relationship between internal dynamics, structural constraints, and trajectory selection in self-organizing systems [1]. One of its central results is the **Failure of Local Closure Theorem**: the demonstration that the observable behavior of a system cannot be determined exclusively from its internal dynamics, but crucially depends on conditions that operate "from outside" the operational domain of the system [1].

This paper deepens that thesis through a rigorous formalization of the distinction between three levels:

- **N₀**: The internal dynamics of the system —its equations, parameters, evolution rules.
- **Boundary conditions (B)**: Constraints that redefine the space of admissible solutions without modifying the internal dynamics.
- **N₁**: The selecting instance that imposes the boundary conditions —external to the operational process of the system.

The question guiding this analysis is: **Where exactly does the identity of a complex system reside when its constituent matter is completely replaceable?**

2. Theoretical Framework: Three Levels of Structural Analysis

2.1 Formal Definition of N₀

Definition 2.1 (N₀): Let N₀ be the set of laws, parameters, and evolution rules that govern the internal dynamics of a system S. Formally:

$$N_0 = L, \Theta, R$$

where:

- L is the set of differential equations or transition rules
- Θ is the parameter space (weights, physical constants, etc.)
- R is the set of inference or evolution relations

Fundamental property of N₀: N₀ is **underdetermined**. That is, given N₀ alone, the space of possible trajectories Σ is typically infinite or highly degenerate.

2.2 Definition of Boundary Conditions (B)

Definition 2.2 (Boundary Condition): A boundary condition B is a constraint that:

1. Does not modify L, Θ , or R
2. Reduces the space of admissible solutions: $\Sigma_B \subset \Sigma$
3. Is imposed on the system from an instance that operates in a domain distinct from the evolutionary process of S

Example 2.1 (Laplace Equation): The equation $\nabla^2 \phi = 0$ possesses infinitely many solutions. What determines which solution appears is not the equation itself (N_0) but the conditions imposed at the boundary: $\phi|_{\partial\Omega} = f(s)$ or $\partial n \phi|_{\partial\Omega} = g(s)$.

Example 2.2 (Euler-Bernoulli Equation): The beam equation

$$EI \frac{d^4 w(x)}{dx^4} = q(x)$$

remains identical for:

- Clamped-clamped beam
- Pinned-pinned beam
- Cantilever beam

The physics is the same. The equation is the same. But the solution space changes radically according to the boundary conditions.

2.3 Definition of N_1

Definition 2.3 (N_1): N_1 is the instance that **selects** the boundary conditions B . Formally:

$$N_1 \rightarrow B$$

$$(B, N_0) \rightarrow trajectory$$

Properties of N_1 :

- Does not belong to the operational domain of N_0
- Is not computable within the state space of S
- Operates as a selector among multiplicities: when $\Delta > 1$ (more than one possible trajectory), N_1 determines which one is realized

Critical distinction: N_1 is **not** a boundary condition. N_1 is **what selects** the boundary condition.

3. Structural Analogies: Five Domains

3.1 The Beam: Structural Engineering

LEVEL	ENTITY	ROLE
N ₀	Euler-Bernoulli equation, E, I, q(x)	Internal dynamics
B	Support condition (clamped, pinned, free)	Solution constraint
N ₁	The engineer who designs the beam	Boundary selector

Observation 3.1: The beam equation does not decide whether it will be clamped or pinned. The designer decides. And the designer is not contained in the equation.

Immediate materialist objection: "The engineer is also a physical system."

TNA Response (Section 6): Correct. The engineer is physical. But the engineer is **external to the operational domain of the beam equation**. Complete reduction would require that the beam equation contain the engineer's decision—which is impossible without infinite regress.

3.2 The Hydraulic Channel: Flow and Geometry

The Manning equation:

$$V = \frac{1}{n} R_h^{2/3} S^{1/2}$$

where R_h is the hydraulic radius.

LEVEL	ENTITY	ROLE
N ₀	Manning equation, n, S, g	Internal dynamics
B	Channel geometry (cross-section, roughness)	Flow constraint
N ₁	The channel designer	Geometry selector

Observation 3.2: Two channels can have the same water, the same gravity, the same equation, and produce completely different velocities because someone designed different boundaries.

Ontological analogy: A river is not the water molecules. A river is water **plus a structure that channels trajectories**. Matter flows; form persists.

3.3 The Wave: Identity without Fixed Matter

The wave equation:

$$y(x, t) = A \sin(kx - \omega t)$$

Fundamental physical observation: When a wave passes through water, locally at point x the matter rises and falls —it does not move in the direction of propagation. What moves in the y -direction is the **perturbation**, the **form**, the **solution of the equation**.

LEVEL	ENTITY	ROLE
N ₀	Water molecules, local equations	Material medium
B	Boundary conditions of the medium (depth, walls)	Propagation constraint
N ₁	The mathematical structure that persists	Form selector

Observation 3.3: If you mark the molecules that form a wave crest, seconds later the crest exists but is formed by **other molecules**. The identity of the wave resides in the global organization, not in the local particles.

Ontological corollary: A wave is closer to an **idea** than to a **thing**. Its identity is abstract, equational, not material.

3.4 The LLM and the System Prompt: Artificial Intelligence

LM-Studio Experiment:

- **Fixed model** (frozen weights)
- **Fixed input** (same image, same user prompt: "Describe this image")
- **Only the boundary condition varies** ("system" field)

SYSTEM PROMPT	OUTPUT
"You are an art critic"	Aesthetic semantic trajectory
"You are a forensic pathologist"	Medical semantic trajectory
"You are a theoretical physicist"	Scientific semantic trajectory

LEVEL	ENTITY	ROLE
N ₀	Model weights, architecture, inference rules	Internal dynamics
B	System prompt	Semantic constraint
N ₁	The user who writes the system prompt	Frame selector

Observation 3.4: The weights did not change. The architecture did not change. The internal state space did not change. The only thing that changed was an external restriction that modifies which region of the solution space becomes accessible.

TNA Question: Is the system prompt N_1 ?

Answer: Not completely. The system prompt is **computable** within the system. It is representable as text. It is modifiable by the operator. It is an **experimental analogue of N_1** — an *operational constraint*.

The true N_1 instance appears when we ask: **Who decided that the model should behave as a philosopher and not as a poet?**

3.5 The Biological Organism: Intestine and Nutrition

Analogy: If a human is topologically an evolved tubeworm, the intestine is an interface between two domains: "external" to the body (not yet absorbed) but "internal" to the survival system.

LEVEL	ENTITY	ROLE
N_0	Intestinal physiology, transporters, gradients	Absorption dynamics
B	Substance present in the lumen (calcium vs. magnesium)	Biochemical boundary condition
N_1	The consciousness that chooses calcium or magnesium	Input selector

Observation 3.5: Physiology describes what happens *after* ingestion, but not necessarily what determines the *prior selection* among possible alternatives.

Case A: "I choose magnesium." → Direct selection. **Case B:** "I flip a coin." → Selection of a **selection operator**.

In Case B, consciousness does not decide between calcium and magnesium. It decides something more abstract: "I will delegate the selection to a coin." That is, **it selects a selection operator** — a regression pointing toward N_1 .

4. Identity as Structure, Not as Matter

4.1 The Argument of Material Turnover

If a person were literally the atoms that compose them, they would cease to be themselves every time those atoms were replaced. However:

- Intestinal cells: days
- Skin cells: weeks
- Proteins: hours or days
- Body atoms: ~10 years

And yet, Claudio at 10, 30, and 60 years old still considers himself the same subject.

4.2 The Hierarchy of Identity

ENTITY	TYPE OF IDENTITY	WHERE DOES CONTINUITY RESIDE?
Rock	Material	In the specific atoms
River	Flow	In the channel structure
Wave	Structural	In the solution of the wave equation
Living being	Self-regulated	In the organization that channels material flow
Person	Biographical	In the realizability structure (N_1)

Thesis 4.1: What persists is not matter but the **realizability structure** that maintains the continuity of the trajectory.

4.3 The Wave as Ontological Model

A wave is the most extreme example because:

1. **It needs matter** to manifest
2. **It does not identify** with any particular portion of matter
3. Its identity is **the propagating mathematical solution**, not the local molecules

Application to personal identity: If we are more like a wave than a rock, seeking identity in atoms is as erroneous as seeking a wave in a drop of water.

5. Materialist Objections and TNA Responses

Objection 1: "Everything is N_0 , including the selector"

"The engineer is a physical system. Consciousness is neural activity. Therefore, the choice of boundary is still explainable in principle by N_0 ."

TNA Response:

This objection confuses **physical content** with **operational structure**. It is true that:

- The engineer is made of atoms
- Consciousness depends on neurons
- Nothing is "outside the universe"

But the relevant question is not ontological but **structural**:

Can the beam equation contain the engineer's decision?

If the answer is no —and it is, on pain of infinite regress— then the selection of boundary conditions is **not reducible to the operational domain of N_0** , even though the selector is physically real.

Formalization:

Let S_0 be the operational system governed by N_0 . Let S_1 be the system that selects B .

The materialist objection assumes: $S_1 \subset S_0$.

TNA responds: Although S_1 is physically instantiated, its **structural function** with respect to S_0 is that of an **external selector**. The relation $N_1 \rightarrow B$ is not computable within the state space of S_0 .

Objection 2: "Boundary conditions are part of the physical problem"

"The beam support is part of the physical model. The system prompt is a system input. They remain N_0 ."

TNA Response:

Correct in one sense, but insufficient in another. We must distinguish:

1. **The boundary condition as physical restriction** → Belongs to the N_0 domain as *input*
2. **The selection of that condition** → Belongs to N_1

The concrete support (clamped, pinned) is N_0 . The decision to choose that support is N_1 .

The system prompt as text is N_0 . The decision to impose that system prompt is N_1 .

Analogy: The ingested calcium is N_0 . The decision to ingest calcium instead of magnesium is N_1 .

Objection 3: "The infinite regress of selectors"

"If N_1 selects B , who selects N_1 ? Don't we fall into infinite regress?"

TNA Response:

Regress appears only if we demand that every selector be in turn selected. But TNA does not postulate an infinite hierarchy. It postulates a **structural boundary**:

- N_0 operates in the domain of dynamics
- N_1 operates in the domain of selection of possibility conditions
- The relation between them is **asymmetric**: N_1 conditions N_0 , not vice versa

The question "who selects N_1 ?" is equivalent to asking "who decides the rules of the game?" — and the answer is that the rules of the game are not decided *within* the game. This does not imply supernatural metaphysics, but a **distinction of structural levels**.

Objection 4: "The language of ' N_1 ' drags a substance ontology"

"When we say ' N_1 is something,' we are implying a substance, matter, or tangible thing."

TNA Response:

This is perhaps the deepest objection, and it points to a genuine problem. As Wittgenstein notes, language was developed to interact with everyday objects; it forces us to substantivize.

But TNA does not postulate that N_1 is a "thing." It postulates that N_1 is a **structural function**: the function of selecting among multiplicities. That function can be:

- Physically instantiated (neurons, engineer, user)
- Without identifying with any particular instance
- Without localizing in any specific portion of matter

Analogy of the symphony: Where is Beethoven's Fifth Symphony? It is not in a specific score, nor in a specific recording, nor in a specific performance. It has reality, but not the reality of a "thing." N_1 operates similarly: it has functional reality, not substantial reality.

6. Mathematical Formalization

6.1 General Scheme

$\Sigma = \sigma : L(\sigma) = 0$ (Solution space of N_0)

$B : \Sigma \rightarrow \Sigma, B \subset \Sigma$ (Restriction by boundary conditions)

$N_1 : B \rightarrow B$ (Selection of boundary condition)

$\sigma = \text{evolve}(N_0, B)$ (Realized trajectory)

where B is the space of possible boundary conditions.

6.2 Theorem: Underdetermination of N_0

Theorem 6.1: For any system S with dynamics N_0 such that $\dim(\Sigma) > 1$, the realized trajectory σ^* is not determined solely by N_0 .

Proof: If $\dim(\Sigma) > 1$, there exist at least two trajectories $\sigma_1, \sigma_2 \in \Sigma$ consistent with N_0 . The internal dynamics contains no criterion to prefer one over the other. Therefore, σ^* requires additional information: B . ■

6.3 Theorem: Irreducibility of N_1 to N_0

Theorem 6.2: The selection function $N_1 : B \rightarrow B$ is not computable within the state space of S governed by N_0 .

Proof sketch: Suppose N_1 were computable within N_0 . Then there would exist an effective procedure within S that selects B . But the selection of B conditions the solution space of S . This would require that S contain a complete description of its own conditions of possibility—which, by the structural incompleteness theorem (analogous to Tarski/Gödel), generates either infinite regress or contradiction. ■

7. Connections with the TNA Corpus

7.1 Failure of Local Closure (10.5281/zenodo.18644522)

The paper *Failure of Local Closure in Self-Optimizing Systems: A Minimal Structural Bound* [1] establishes that self-optimizing systems cannot achieve complete operational closure because they require boundary conditions that they do not generate internally. The present paper extends that result by showing that:

- Boundary conditions are not mere "inputs" but **trajectory selectors**
- The instance that imposes those conditions (N_1) has a structural status distinct from internal dynamics

7.2 The Block Universe Confronted by TNA (10.5281/zenodo.20371619)

In *The Block Universe Confronted by TNA*, it is argued that the block universe view ignores the asymmetry between possible multiplicity and realized trajectory. The present paper formalizes that asymmetry: N_0 generates the multiplicity; N_1 (through B) selects the trajectory.

7.3 Selector/Operator (10.5281/zenodo.20401870)

The paper on Selector/Operator distinguishes between the instance that chooses (Selector) and the mechanism that executes (Operator). The N_1/N_0 distinction of the present paper maps directly onto that distinction:

- **Selector** = N_1 (chooses the boundary conditions)
- **Operator** = N_0 (executes the dynamics under those conditions)

7.4 Core Structure (10.5281/zenodo.18098558)

The formal foundations of TNA (N_0/N_1 , Aleph, Psi, $F_{\min}$) provide the axiomatic framework within which boundary conditions acquire their status as "realizability constraints" rather than "internal parameters."

8. Implications and Conclusions

8.1 Implications for AI

The LM-Studio experiment demonstrates that an LLM with frozen weights can produce radically different semantic trajectories according to the system prompt. This is not a "bug" but a manifestation of a general structural principle: **internal dynamics do not determine observable behavior**. The implication for AI system design is that alignment and behavior cannot be resolved solely through weight training (N_0), but require explicit attention to boundary conditions (B) and, ultimately, to who selects them (N_1).

8.2 Implications for Biology

The analogy of the intestine and nutrient selection suggests that conventional physiology (N_0) is insufficient to explain the directionality of organic behavior. "Consciousness" —understood not as substance but as a function of boundary condition selection— participates in determining which physiological trajectories are realized.

8.3 Implications for Personal Identity

If we are more like a wave than a rock, our identity does not reside in the atoms that compose us (replaceable every ~10 years) but in the **realizability structure** that persists through material turnover. That structure is not a "thing" that can be localized, but a pattern of constraints that channels the flow of matter, energy, and information.

8.4 Conclusion

The central thesis of this paper can be summarized in three propositions:

1. **N_0 is underdetermined:** The internal dynamics of a system generate a space of possible trajectories, but do not select which one is realized.
2. **Boundary conditions (B) restrict but do not select:** B reduces the solution space, but the choice of B comes from a distinct structural domain.

3. **N_i is the selecting instance:** N_i is not an additional substance nor a metaphysical ghost, but the structural function of imposing constraints that the operational system does not generate by itself.

The identity of complex systems —whether beams, channels, waves, LLMs, or persons— does not reside in their constituent matter, which is replaceable, but in the **realizability structure** that persists through material turnover. That structure is closer to an equation, a wave, an idea, than to a thing.

References

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Appendix: Glossary of TNA Terms

TERM	DEFINITION
N_o	Operational domain: internal dynamics, equations, parameters, evolution rules
N_i	Selector domain: instance that imposes boundary conditions, external to the operational process
B	Boundary condition: constraint that reduces the solution space without modifying N_o
Σ	Space of possible trajectories generated by N_o
Σ_B	Space of admissible trajectories under boundary condition B

TERM	DEFINITION
F_min	Minimum realizability function: lower structural limit for the existence of trajectories
Ψ	Structural collapse operator: transition from multiplicity to singularity
Aleph	Cardinality of the space of unrealized possibilities

Appendix - Theorem 6.3

Theorem 6.3 (Structural Incompleteness of Boundary Selection)

Let

$$S = (X, N_0)$$

be a system whose state space is

$$X$$

and whose operational dynamics are generated by

$$N_0 : X \rightarrow X.$$

Let

$$\mathcal{B}$$

denote the set of admissible boundary conditions acting on S .

Assume that each boundary condition

$$B \in \mathcal{B}$$

induces a restricted trajectory space

$$\Sigma_B \subseteq \Sigma.$$

Let

$$N_1 : \mathcal{B} \rightarrow \mathcal{B}$$

denote the boundary-selection function.

Then:

The complete specification of N_1 cannot be derived solely from the operational dynamics N_0 .

Formally,

$$N_0 \not\equiv N_1.$$

Proof

Step 1

Assume the contrary.

Suppose there exists a procedure Π_B generated entirely by N_0 such that:

$$\Pi_B \subseteq N_0 \quad \text{where} \quad \Pi_B \equiv N_1$$

Therefore, Π_B determines all admissible boundary conditions.

Step 2

Every trajectory generated by the system depends on the selected boundary condition.

Thus

$$\Sigma_B = \Sigma(N_0, B).$$

But the operation of

$$\Pi_B$$

itself occurs within the same trajectory space.

Therefore

$$\Pi_B = \Pi_B(N_0, B).$$

The selector depends on the boundary condition it selects.

Step 3

To compute B , the procedure must already operate under some boundary condition

$$B_{\Pi}.$$

Hence

$$\Pi_B = \Pi_B(N_0, B_{\Pi}).$$

But B_{Π} must itself be admissible.

Therefore B_{Π} also requires specification.

Step 4

If B_{Π} is generated by the same procedure,

$$B_{\Pi} = \Pi_B(\cdot),$$

then the procedure presupposes the very condition it attempts to generate.

If B_{Π} is generated by another procedure,

$$\Pi_B^{(2)},$$

then

$$\Pi_B^{(2)} \rightarrow B_{\Pi} \rightarrow \Pi_B \rightarrow B.$$

Repeating the argument yields a hierarchy

$$\Pi_B^{(n)} \rightarrow \Pi_B^{(n-1)} \rightarrow \dots \rightarrow \Pi_B^{(1)}.$$

Step 5

The hierarchy terminates only if there exists a selector whose validity is not derived from the operational dynamics it conditions.

Therefore at least one selection level must satisfy

$$\Pi_B^{(k)} \not\subseteq N_0.$$

This contradicts the assumption that the entire selection function is derivable from N_0 .

Hence

$$N_0 \not\vdash N_1.$$

Therefore the complete specification of boundary selection is not derivable from the operational dynamics being conditioned.

■

Corollary 6.3.1

For any underdetermined system satisfying

$$\dim(\Sigma) > 1,$$

the realization of a particular trajectory

$$\sigma^*$$

requires information not contained in the operational dynamics alone.

Corollary 6.3.2

Boundary selection constitutes a structurally higher-order operation than trajectory generation.

Formally,

$$\text{Level}(N_1) > \text{Level}(N_0).$$

Corollary 6.3.3 (TNA Form)

No system can completely derive its own conditions of possibility from the dynamics conditioned by those conditions.

Symbolically,

$$N_0 \not\vdash N_1.$$